

QA LEADERSHIP ACADEMY

Automation Framework Decision Matrix

Choose a tooling direction on evidence and weighted criteria, not on whichever tool the loudest engineer prefers.

Purpose

This is a vendor-neutral, weighted decision instrument for selecting or changing an automation framework. It forces the criteria and their weights to be agreed before any tool is scored, so the choice reflects your context — team skills, application architecture, feedback-speed needs — rather than market hype or personal preference.

When to use it

- When an inherited framework is holding you back and a re-platform is on the table.
- When starting automation from scratch for a new product or team.
- When two squads disagree on tooling and you need an objective, shared decision.

Instructions

1. Agree the criteria and weights with your stakeholders first, before naming any tool — this is the step that removes bias.
2. Shortlist two or three genuine options (describe them by capability, e.g. "current UI tool", "modern browser-automation tool", "API-first tooling"), plus the status quo as a baseline.
3. Score each option 1-5 per criterion against evidence — ideally a small spike, not a datasheet.
4. Multiply by weight, total each column, and read the leader alongside the migration cost.
5. Sanity-check the winner against your biggest constraint (usually team skills and feedback speed) before committing.

Worked example

Northstar is weighing whether to keep its Selenium UI suite or shift the centre of gravity. The team is small (6 QA, one automation-leaning engineer), feedback speed is a top pain (~6-hour run) and the app is a React SPA plus REST APIs, so API-level testing is weighted heavily. Options are described by capability, not brand:

Criterion	Weight	Keep current UI suite	Modern browser tool	API-first + thin UI
Team skills / ramp-up	20	4	2	3

Criterion	Weight	Keep current UI suite	Modern browser tool	API-first + thin UI
Feedback speed	20	1	3	5
Reliability / low flake	20	1	3	5
Fit to architecture (SPA + REST)	15	2	4	5
Maintenance cost	15	2	3	4
Migration cost (higher = better)	10	5	2	2
Weighted total (out of 100)	100	38	57	81

The status quo wins only on migration cost — unsurprising, since doing nothing is always cheapest today. The API-first direction, with a thin UI layer for a few critical journeys, scores highest because it attacks Northstar's two worst pains (feedback speed and flakiness) and fits the REST architecture. Note the honest trade-off: it carries a real migration cost, which is why the recommendation is an incremental shift, not a big-bang rewrite.

WEIGHT BEFORE YOU SCORE

The order matters. Agree the weights while the options are still anonymous. Once a favourite tool is named, people unconsciously weight the criteria to make it win.

Blank reusable version

Criterion	Weight	Option A	Option B	Option C
Team skills / ramp-up				
Feedback speed				
Reliability / low flake				
Fit to architecture				
Maintenance cost				
Ecosystem / longevity				
Migration cost (higher = better)				
Weighted total	100			

Common mistakes

COMMON MISTAKES

Scoring tools before agreeing weights, so the numbers rationalise a decision already made; comparing only new tools and omitting the status quo as a baseline; scoring from datasheets rather than a short spike; ignoring team ramp-up and treating the framework choice as purely technical; and understating migration cost so the winner looks free.

How to interpret the results

- A narrow win (a few points) is not a mandate — reduce it to the one or two criteria that decide it and confirm those hold.
- If the status quo only wins on migration cost, that is a classic signal that inertia, not fit, is keeping you where you are.
- Read the winner against your heaviest-weighted criterion; a tool that loses on your top pain is the wrong winner even with a high total.
- Pair the decision with a migration approach (incremental vs big-bang); the matrix chooses direction, not sequencing.

INSIDE STLC PRO TIP

Keep this matrix vendor-neutral by scoring capabilities, not brands, until the final step. When you do present it, lead with the weights you agreed together — a decision your stakeholders helped weight is a decision they will defend rather than second-guess.